

Experiment No. 1

Setting up Hyperledger Fabric Development Environment

AIM:

To install and configure the Hyperledger Fabric development environment on a Linux system using Docker, Go, and Hyperledger Fabric binaries, create a blockchain test network, and establish a communication channel between peer organizations.

SOFTWARE REQUIREMENTS:

- Kali Linux / Ubuntu
- Docker
- Docker Compose
- Go (Version 1.20 or later)
- Git
- Curl
- Hyperledger Fabric Samples
- Hyperledger Fabric Docker Images

PROCEDURE:

Step 1: Update the System

```
sudo apt update
```

```
sudo apt upgrade -y
```

Step 2: Install Required Packages

```
sudo apt install -y git curl jq docker-compose golang-go
```

```
git --version
```

```
curl --version
```

```
docker-compose --version
```

```
go version
```

Step 3: Verify Docker Installation

Start Docker:

```
sudo systemctl start docker
```

```
sudo systemctl enable docker
```

Add current user:

```
sudo usermod -aG docker $USER
```

```
newgrp docker
```

Verify Docker:

```
docker run hello-world
```

Step 4: Create Workspace

```
mkdir -p ~/fabric-dev
```

```
cd ~/fabric-dev
```

```
pwd
```

Step 5: Download Hyperledger Fabric

```
curl -sSL https://bit.ly/2ysbOFE | bash -s
```

Step 6: Navigate to Fabric Samples

```
cd fabric-samples
```

Step 7: Configure Environment Variables

```
echo 'export PATH=$PATH:$HOME/fabric-dev/fabric-samples/bin' >> ~/.bashrc
```

```
source ~/.bashrc
```

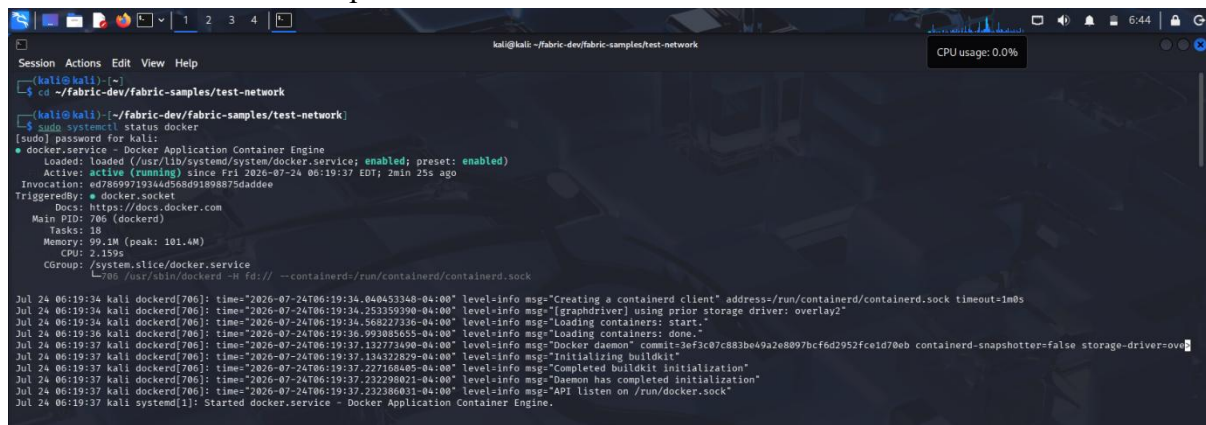
Step 8: Verify Hyperledger Fabric

```
peer version
```

```
orderer version
```

Step 9: Start the Test Network

```
cd ~/fabric-dev/fabric-samples/test-network
```



```
kali@kali: ~/fabric-dev/fabric-samples/test-network
CPU usage: 0.0%

(kali@kali)-[~/fabric-dev/fabric-samples/test-network]
$ cd ~/fabric-dev/fabric-samples/test-network
(kali@kali)-[~/fabric-dev/fabric-samples/test-network]
$ sudo systemctl status docker
[sudo] password for kali:
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-07-24 06:19:37 EDT; 2min 25s ago
   Invocation: e07869729244d5b8d91898875daddee
   TriggeredBy: ● docker.socket
   Docs: https://docs.docker.com
   Main PID: 786 (dockerd)
   Tasks: 18
   Memory: 99.1M (peak: 101.4M)
   CPU: 2.15s
   CGroup: /system.slice/docker.service
           └─/usr/sbin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Jul 24 06:19:34 kali dockerd[786]: time="2026-07-24T06:19:34.040453748-04:00" level=info msg="Creating a containerd client" address=/run/containerd/containerd.sock timeout=1m0s
Jul 24 06:19:34 kali dockerd[786]: time="2026-07-24T06:19:34.253359290-04:00" level=info msg="[graphdriver] using prior storage driver: overlay2"
Jul 24 06:19:34 kali dockerd[786]: time="2026-07-24T06:19:34.568227336-04:00" level=info msg="Loading containers: start."
Jul 24 06:19:36 kali dockerd[786]: time="2026-07-24T06:19:36.993895655-04:00" level=info msg="Loading containers: done."
Jul 24 06:19:37 kali dockerd[786]: time="2026-07-24T06:19:37.132773490-04:00" level=info msg="Docker daemon" commit=3ef3c07c883b649a2e8097bcfd2952fcd70eb containerd-snapshotter=false storage-driver=overlay2
Jul 24 06:19:37 kali dockerd[786]: time="2026-07-24T06:19:37.134322829-04:00" level=info msg="Initializing buildkit"
Jul 24 06:19:37 kali dockerd[786]: time="2026-07-24T06:19:37.227168405-04:00" level=info msg="Completed buildkit initialization"
Jul 24 06:19:37 kali dockerd[786]: time="2026-07-24T06:19:37.232298021-04:00" level=info msg="Daemon has completed initialization"
Jul 24 06:19:37 kali dockerd[786]: time="2026-07-24T06:19:37.232386031-04:00" level=info msg="API listen on /run/docker.sock"
Jul 24 06:19:37 kali systemd[1]: Started docker.service - Docker Application Container Engine.
```

```
./network.sh up
```

```
(kali@kali) ~ - /fabric-dev/fabric-samples/test-network
$ ./network.sh up
Using docker and docker-compose
Starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database 'leveldb'
LOCAL_VERSION=v2.5.16
DOCKER_IMAGE_VERSION=v2.5.16
[*] Running 3/3
  Container peer0.org2.example.com Started 1.0s
  Container orderer.example.com Started 1.6s
  Container peer0.org1.example.com Started 1.1s
CONTAINER ID        STATUS      PORTS
e4b442874c07      dev-peer0.org1.example.com-basic.1.0-1f66cc610fb8ac88d69e78edc4c1cb09bac7c839fb5399009310ac8dfa703799-8f4296b79b03dc5e553784fa18a614f0c6e4174be802b7338869e17a94c1367 "chaincode -peer.add...
dev-peer0.org1.examl
18 minutes ago      Exited (2) 3 minutes ago
e.com-basic.1.0-1f66cc610fb8ac88d69e78edc4c1cb09bac7c839fb5399009310ac8dfa703799 "chaincode -peer.add...
533acc3fcfd4      dev-peer0.org2.example.com-basic.1.0-1f66cc610fb8ac88d69e78edc4c1cb09bac7c839fb5399009310ac8dfa703799-77fa314a0214359a817dc282f148d1d72ce27da08c59102dc659741e9ed3349f "chaincode -peer.add...
dev-peer0.org2.examl
18 minutes ago      Exited (2) 3 minutes ago
e.com-basic.1.0-1f66cc610fb8ac88d69e78edc4c1cb09bac7c839fb5399009310ac8dfa703799 "chaincode -peer.add...
f3bfa331178b      40bab10fe51d " /bin/sh -c 'M(nop) -
focused_nash
05a2e99f7973      a07dae9f0d80 " /bin/sh -c 'M(nop) -
agitated_wiles
aedf6daf304d      9c00230ef129 " /bin/sh -c 'M(nop) -
bold_blackwell
c0f02c6374c9      39222153a41f " /bin/sh -c 'M(nop) -
jolly_wu
37f638ea5803      hyperledger/fabric-peer:latest "peer node start"
peer0.org1.example.co
22 minutes ago      Up 1 second      0.0.0.0:7051->7051/tcp, [::]:7051->7051/tcp, 0.0.0.0:9444->9444/tcp, [::]:9444->9444/tcp
```

Step 10: Verify Running Containers

docker ps

Step 11: Create Channel

./network.sh createChannel -c mychannel

```
(kali@kali) ~ - /fabric-dev/fabric-samples/test-network
$ ./network.sh createChannel -c mychannel
Using docker and docker-compose
Creating channel 'mychannel'
If network is not up, starting nodes with CLI timeout of '3' tries and CLI delay of '3' seconds and using database 'leveldb'
LOCAL_VERSION=v2.5.16
DOCKER_IMAGE_VERSION=v2.5.16
[*] Running 3/3
  Container peer0.org2.example.com Running 0.0s
  Container orderer.example.com Running 0.0s
  Container peer0.org1.example.com Running 0.0s
CONTAINER ID        STATUS      PORTS
4b0a9d0c0c0a      dev-peer0.org1.example.com-basic.1.0-1f66cc610fb8ac88d69e78edc4c1cb09bac7c839fb5399009310ac8dfa703799-77fa314a0214359a817dc282f148d1d72ce27da08c59102dc659741e9ed3349f "chaincode -peer.add...
dev-peer0.org1.examl
6 seconds ago      Up 4 seconds
com-basic.1.0-1f66cc610fb8ac88d69e78edc4c1cb09bac7c839fb5399009310ac8dfa703799 "chaincode -peer.add...
dev-peer0.org1.examl
0 seconds ago      Up 4 seconds
dev-peer0.org1.examl
18 minutes ago      Created
f3bfa331178b      40bab10fe51d " /bin/sh -c 'M(nop) -
focused_nash
05a2e99f7973      a07dae9f0d80 " /bin/sh -c 'M(nop) -
agitated_wiles
aedf6daf304d      9c00230ef129 " /bin/sh -c 'M(nop) -
bold_blackwell
c0f02c6374c9      39222153a41f " /bin/sh -c 'M(nop) -
jolly_wu
37f638ea5803      hyperledger/fabric-peer:latest "peer node start"
peer0.org1.example.co
22 minutes ago      Up 12 seconds      0.0.0.0:7051->7051/tcp, [::]:7051->7051/tcp, 0.0.0.0:9444->9444/tcp, [::]:9444->9444/tcp
dev-peer0.org1.examl
22 minutes ago      Up 12 seconds      0.0.0.0:7050->7050/tcp, [::]:7050->7050/tcp, 0.0.0.0:7051->7051/tcp, [::]:7051->7051/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp
b12b0a261e19      hyperledger/fabric-peer:latest "peer node start"
peer0.org2.example.co
22 minutes ago      Up 12 seconds      0.0.0.0:9001->9001/tcp, [::]:9001->9001/tcp, 7051/tcp, 0.0.0.0:9445->9445/tcp, [::]:9445->9445/tcp
5aa0a9292dad      hello-world "hello-world"
6 hours ago        Exited (0) 6 hours ago
3347c7cf78ba      hello-world "hello-world"
6 hours ago        Exited (0) 6 hours ago
Using docker and docker-compose
Generating channel genesis block 'mychannel.block'
Using application configuration
/home/kali/fabric-dev/fabric-samples/test-network/...bin/configtxgen
+ '[' 0 == 1 ']'
+ configtxgen -profile ChannelUsingBaf -outputBlock ./channel-artifacts/mychannel.block -channelID mychannel
2020-07-24 06:12:15.419 EDT 0001 INFO [common.tools.configtxgen] main -> Loading configuration
2020-07-24 06:12:15.420 EDT 0002 INFO [common.tools.configtxgen.localconfig] completeInitialization -> orderer type: etcdraft
2020-07-24 06:12:15.468 EDT 0003 INFO [common.tools.configtxgen.localconfig] completeInitialization -> Orderer.StcBaf.Options unset, setting to tick_interval: "500ms" election_tick:10 heartbeat_tick:1 max
+ flight_blocks15 snapshot_interval_size:16777216
2020-07-24 06:12:15.469 EDT 0004 INFO [common.tools.configtxgen.localconfig] Load -> Loaded configuration: /home/kali/fabric-dev/fabric-samples/test-network/configtx/configtx.yaml
2020-07-24 06:12:15.518 EDT 0005 INFO [common.tools.configtxgen] doOutputBlock -> Generating genesis block
2020-07-24 06:12:15.518 EDT 0006 INFO [common.tools.configtxgen] doOutputBlock -> Creating application channel genesis block
2020-07-24 06:12:15.519 EDT 0007 INFO [common.tools.configtxgen] doOutputBlock -> Writing genesis block
+ res0
Creating channel 'mychannel'
Adding nodes
+ ./scripts/orderer.sh mychannel
+ '[' 0 == 1 ']'
+ res0
```

Step 12: Verify Channel

